

Abhishek Mule

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Professional Summary

I am an adaptable Electronics and Telecommunication Engineering student with hands-on experience in embedded systems, STM32 firmware development, PCB design, and safety-critical electronics through Formula Student Electric Vehicle projects. Experienced in hardware testing, debugging, system validation, troubleshooting, and embedded firmware development with strong analytical and problem-solving skills. Passionate about developing reliable engineering solutions and diagnostic systems through collaborative product development.

Education

Bachelor of Technology in Electronics and Telecommunication

August 2023 - Present

Honors in Semiconductor Technology

June 2025 - Present

Pimpri Chinchwad College of Engineering (PCCOE), Affiliated to Savitribai Phule Pune University, India

Technical Skills

Programming Languages : Embedded C

Development Tools : STM32 CubeIDE, LTSpice, Arduino IDE, Altium, MATLAB.

Equipment : Multimeter, Oscilloscope, Function Generator, DC Load.

Microcontrollers : STM32, Arduino Series, ESP Series.

Communication Protocols : CAN, I2C, SPI, UART.

Debug & Validation : Hardware Testing, Debugging, System Validation, Software-in-the-Loop Testing, Hardware Troubleshooting.

Engineering : Root cause analysis, Failure analysis, Documentation, Technical Troubleshooting, Problem solving, Analytical thinking.

Relevant Engineering Projects

Real-time environment monitoring system using ESP & FreeRTOS

Tech Stack: ESP32, FreeRTOS, Embedded C, Wi-Fi, Web Server.

- Developed a real-time environmental monitoring system using ESP32 and FreeRTOS, integrating DHT11 and MQ135 sensors for monitoring temperature, humidity, and air quality.
- Implemented multitasking using FreeRTOS, managing sensor data acquisition, OLED display updates, and Wi-Fi-based web server communication simultaneously.
- Developed and tested an embedded monitoring system using ESP32 and FreeRTOS, debugging sensor interfaces and validating real-time system performance through functional testing.

Tractive System Active Light

Tech Stack: Analog Circuit Design, PCB Design, Altium, LTSpice.

- Designed, assembled, tested, and validated a Tractive System Active Light PCB complying with Formula Student EV safety regulations.
- Implemented logic such that the light blinks red when tractive system voltage exceeds 60V, turns green when voltage is below 60V, and enters a safe state (no light) during fault conditions.
- Replaced the earlier logic-gate-based design with an op-amp and 555 IC based 2-layer PCB circuit, improving response speed by 15% and reducing PCB size by 25%.

Brake System Plausibility Device

Tech Stack: Analog Circuit Design, PCB Design, Altium, LTSpice.

- Designed, assembled, tested, and validated a Brake System Plausibility Device that monitored simultaneous brake and accelerator inputs, automatically cutting motor power when motor output exceeded 5 kW and brake pressure exceeded 30 bar to ensure driver safety.
- Replaced the earlier logic-gate-based design with an op-amp and MOSFET based circuit, ensuring faster response and improved reliability in safety-critical conditions.

- Reduced PCB size by 35% and improved response speed by 18%, while simplifying circuit complexity and ensuring the system entered a safe state during any failure condition.

Battery Management System for Electric Vehicle

Tech Stack: PCB Design, Altium, ISO-SPI Communication, EV Battery Systems.

- Designed, assembled, tested and validated a 8-layer BMS slave PCB, with 2 boards per module, ensuring reliable cell monitoring and communication with the BMS Master through ISO-SPI interface.
- Implemented voltage sensing, temperature monitoring, and passive cell balancing functions for each board to ensure battery safety, performance, and reliable operation.
- Collaborated with multidisciplinary electrical, mechanical, and software teams during PCB development, testing, and system integration.
- Investigated system faults by analyzing sensor signals, communication interfaces, and embedded firmware behavior to improve system reliability.

Electronic Control Unit for Electric Vehicle

Tech Stack: STM32, Embedded C, HAL Library, Sensor Integration, SIL Testing.

- Developed embedded firmware for an STM32-based Electronic Control Unit (ECU), implementing CAN, UART, SPI, and RS232 communication, integrating multiple vehicle sensors, debugging hardware-software interfaces, and validating control algorithms through Software-in-the-Loop (SIL) testing
- Integrated and tested IMU, brake, steering, and wheel-speed sensors while validating control logic through Software-in-the-Loop (SIL) testing.
- Hands-on experience in developing embedded systems that support fault detection, debugging, and troubleshooting of electronic systems.

Certifications & Achievements

- National Champion at Formula Bharat 2025 in the Electric Vehicle category.
- Certified EV Safety Level 3E (SkillShark / ASDC) to work on live high-voltage.
- Certification of online Internship in Internet of Things at Emertex from 2nd March 2025 to 20th April 2025.